UAV-Based Multispectral Object Tracking With Positive–Negative Prompt Mining Network

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***Abstract*—Multispectral imaging (MSI) based on uncrewed aerial vehicle (UAV) can capture valuable spectral, thereby improving the capability of object tracking. However, most existing multispectral (MS) trackers focus on feature matching of the foreground target, leading to target loss due to sparse features and strong interference under complex backgrounds. To address this challenge, we propose the positive–negative prompt mining network (PNPM-Net), which models both background interference and target features as structured negative and positive prompts to enhance feature discriminability under complex scenarios. It mainly consists of three key components: spatial–spectral–temporal attention mining (SSTAM), positive–negative joint mining (PNJM) module, and dual-incentive prompt modulation (DIPM) module. First, we propose SSTAM for adaptive spectral channel reweighting and spatial reshaping, thereby augmenting the spatial–spectral separability between target and background signatures under complex dynamic scenarios. Second, we introduce PNJM to generate contrastive priors modeling the positive and negative prompts. Finally, we develop DIPM to employ these priors for feature enhancement and background suppression, thereby increasing the interclass discriminativeness between target and background. Experiments show PNPM-Net achieves 71.8% AUC and 88.1% DP@20, surpassing state-of-the-art (SOTA) methods by 18.5% and 9.2%, respectively, which demonstrate the strong potential in UAV-based MS object tracking (MOT) area.**

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I. Introduction

VISUAL object tracking based on uncrewed aerial vehicles (UAVs) is a core technology in modern remote sensing [1], [2], [3], supporting critical Earth observation applications, such as border surveillance [4], [5], [6], [7], precision agriculture [8], [9], [10], [11], semantic segmentation [11], [12], [13], and postdisaster assessment [14], [15]. As a flexible remote sensing platform, UAVs provide high-spatiotemporal resolution along user-defined flight paths, bridging the gap between wide-area satellite coverage, and ground-based measurements [16], [17]. Consequently, UAV-based object tracking has become indispensable for dynamic remote sensing tasks. Most existing UAV trackers rely on RGB imagery [18], [19], localizing targets by matching a template with a search region through deep similarity models [20]. However, aerial tracking faces challenges including abrupt viewpoint changes, large scale variations (SVs), motion blur (MB), and cluttered backgrounds [21]. Therefore, these challenges limit the development of UAV-based RGB tracking [22].

To address these limitations, multispectral object tracking (MOT) has emerged as a promising technology for visual tracking [23], [24]. Unlike RGB images that capture only three color channels, multispectral image (MSI) continuously records reflectance across several to dozens of continuous spectral bands [25], [26], thereby displaying the detailed spectral characteristics and intrinsic material properties of object [27], [28]. MOT has evolved along two main technical pathways including handcrafted feature-based methods and deep-learning-based methods [29]. Handcrafted feature-based methods, such as MHT [30], TASSCF [31], and TSCFW [32], manually construct discriminative spatial–spectral descriptors, establishing a foundational principle for modeling target distinguishability, and employ correlation filters or statistical models for object tracking. In contrast, deep learning methods, such as SiamHYPER [33], SEE-Net [34], HCSMP [35], HDSP [36], SSTtrack [37], and UNTrack [26], employ convolutional neural networks or Transformer architectures to automatically extract multilevel joint spatial–spectral representations through end-to-end learning. Compared with manual feature-based

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MOT methods, deep learning frameworks offer the ability to learn features automatically [38]. With their superior capacity for representing complex nonlinear patterns, these frameworks can more effectively capture the discriminative structure and contextual relationships of targets, thereby significantly enhancing the tracking accuracy and robustness in the field of MOT [39].

However, advancing UAV-based MOT remains challenging due to the unique imaging conditions of airborne platforms. Unlike traditional ground-level camera settings, where backgrounds are relatively stable, UAV platforms exhibit rapid ego-motion and frequent attitude changes [40], [41]. This causes drastic appearance variations and often renders targets as small objects with limited pixels [42], [43], [44], leading to sparse feature representations and severe susceptibility to background clutter (BC) interference. Although recent trackers like SSTrack [37] incorporate temporal features, they predominantly adhere to a foreground-centric paradigm that treats background as passive interference. It is worth noting that while some existing methods, such as UNTrack [26], employ background suppression techniques, these mechanisms typically operate by passively filtering out background regions through weight reduction, treating background merely as noise to be ignored rather than a source of discriminative information. This limitation restricts the exploitation of rich spatial–spectral contrast cues embedded in the background, which hinders effective target–background discrimination and exacerbates feature ambiguity, ultimately causing tracking failure under complex dynamic conditions [26], [45].

To address the limitation of foreground-centric paradigms that treat background as passive interference, we propose the positive–negative prompt mining network (PNPM-Net). Distinct from traditional background suppression techniques that passively filter background regions via spatial masking or merely enhance target representation, PNPM-Net introduces a background-aware prompting strategy that actively transforms background interference into discriminative guidance. By explicitly modeling background context as structured negative prompts and jointly leveraging them with target-aware positive prompts, our approach achieves the robust feature-level modulation.

Specifically, the proposed method is organized into three synergistic components to achieve effective spatial–spectral–temporal modeling and background-aware tracking. First, to mitigate spectral ambiguity inherent in complex UAV scenes, the spatial–spectral–temporal attention mining (SSTAM) module performs adaptive spectral channel re-weighting and spatial feature reshaping. Unlike static preprocessing schemes, SSTAM dynamically fuses multitemporal templates to enhance discriminative bands and suppress noise, ensuring robustness against rapid ego-motion and attitude changes. By enhancing the feature signal-to-noise ratio (SNR) and providing spatial constraints, SSTAM establishes a refined foundation that enables the accurate prompt extraction. Second, building on this optimized representation, the positive–negative joint mining (PNJM) module transforms background noise into discriminative signals by generating contrastive priors that model the differences between the target and the background. This mechanism is distinct from existing prompt-based trackers as it actively constructs background-aware negative prompts rather than ignoring them, providing structured priors to overcome foreground-centric limitations. Finally, the dual-incentive prompt modulation (DIPM) module injects these generated prompts into multiple backbone stages. This bidirectional modulation ensures that target responses are enhanced precisely, where BC is attenuated, creating a synergistic effect that sharpens feature contrast more effectively, thereby jointly refining the feature representation.

Through the collaborative optimization of these components, PNPM-Net effectively addresses the challenges of target feature sparsity and background interference in complex UAV scenes. The core innovation of our proposed method centers on the pioneering introduction of the unified background-aware prompting to the MOT field, which shifts the tracking paradigm from passive background suppression to active background utilization, achieving the robust performance in complex scenarios. Extensive experiments demonstrate that PNPM-Net outperforms existing state-of-the-art (SOTA) methods, achieving 71.8% AUC and 88.1% DP@20 with significant improvements of 18.5% and 9.2% over the previous best multispectral (MS) tracker, highlighting its strong potential for robust UAV-based MOT. In summary, PNPM-Net makes the following three contributions.

1) We propose a novel framework for UAV-based MS tracking, termed PNPM-Net, which first explicitly models background context as structured negative prompts for robust tracking in complex scenarios under target sparsity.
2) We design the collaborative SSTAM and PNJM modules to jointly mitigate feature ambiguity via adaptive spectral channel reweighting and spatial reshaping, while generating contrastive prompts to provide the structured positive and negative priors in complex scenarios.
3) We propose the DIPM module, which utilizes contrastive prompting in the form of online-updated parallel modulation to amplify target responses and suppress background regions, thereby enhancing discriminative capabilities in complex scenarios.

II. Related Works

A. MS Object Tracking

MS imaging (MSI) provides much higher spectral resolution than conventional RGB imaging [46], [47], [48]. This allows it to distinguish targets from nontarget regions more effectively, especially when they have different material properties. The advantage is particularly valuable under complex background conditions [22], [49], [50]. Therefore, MSI has attracted increasing attention in the field of visual object tracking [51], [52].

Early study attempts to leverage the discriminative potential of MSI typically relied on handcrafted spatial–spectral features [53], [54], which were then integrated into classical tracking frameworks, such as correlation filters [55], [56]. For instance,

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MHT [30] combines a spectral–spatial histogram of gradients (SSHOG) with material abundance features to achieve material-aware tracking. TASSCF [31] introduced an adaptive spatial–spectral discriminant analysis method that embeds high-dimensional MSI data into a more discriminative low-dimensional subspace. TSCFW [32] leveraged tensor-based processing to model the multidimensional structure of the MS cube to suppress redundant spectral variations in homogeneous backgrounds. SSTCF [46] unified spatial–spectral–temporal constraints within a correlation filter framework and improved temporal consistency across consecutive frames. While these early approaches demonstrated notable efficacy by leveraging handcrafted spatial–spectral descriptors, their inherent reliance on shallow feature representations ultimately constrains their robustness in complex dynamic environments [57].

Recently, researchers have increasingly explored end-to-end architectures such as CNNs and Transformers to fully exploit spatial and spectral information in MS video (MSV) [58]. Early research efforts focused on directly learning informative spectral representations from MSV [33]. For example, SEE-Net [34] utilizes a spectral self-expression model to capture band dependencies and assign importance, constructing pseudo-color images for learning joint spatial–spectral representations and enabling robust tracking through weighted ensemble predictions. Trans-HST [59] introduces a cross-band feature enhancement module to reconstruct and fuse the inherent deep spatial–spectral structure. SPIRIT [60] employs a spectral-aware module to learn band contributions and enhance feature interaction, thereby enabling robust and efficient MS tracking. PHTrack [24] utilizes a modality prompter (MOP) to capture spatial and spectral cues, thereby enhancing spatial–spectral representation knowledge. UBSTrack [57] employs a local–global attention mechanism to dynamically mine key spectral information from MS data, constructing a compact yet effective spectral representation that enhances tracking robustness in complex scenarios. SSTtrack [37] bridges band gaps with a multimodal generative adapter (MGA) to enhance physically interpretable spectral learning, thereby improving the target discrimination capability. HyA-T [27] introduces lightweight adapters that inject spatial and spectral information directly into the self-attention and MLP modules of pretrained RGB trackers, enabling the efficient domain adaptation.

Although the aforementioned methods have significantly enhanced tracking performance, their design paradigms still exhibit notable limitations. Specifically, these methods primarily focus on modeling the spatial–spectral representation of foreground targets while neglecting the discriminative potential of background environments. Consequently, the insufficient utilization of background information collectively restricts the reliability and generalization capability of MOT in UAV-based platform applications.

B. Visual Prompt Learning (VPL)

VPL is a parameter-efficient adaptation technique that incorporates a small set of learnable prompt vectors into pretrained backbone networks to guide feature extraction for specific tasks. In contrast, conventional fine-tuning in visual object tracking adapts a pretrained model by updating its parameters on a target dataset [61], which requires abundant high-quality labeled data. However, such data remain currently unavailable in MOT due to inherent data limitations [62]. To address this scarcity, VPL has emerged as an effective alternative. By guiding the model to integrate task-relevant features with minimal learnable parameters, it can rapidly adapt to new tasks even under limited supervision.

Recent research has advanced the development of MOT through various prompt-based designs. For instance, PHTrack [24] introduces an MOP to capture rich spectral features and bridge interband gaps; HDSP [36] employs a density-based band selection module and a multibranch dual-stream visual prompter (MDVP) to fuse spectral and spatiotemporal cues; HCSMP [35] proposes a cross-modal prompting network with a memory-augmented mechanism to build robust target representations from historical frames. Despite their success, these methods mainly focus on foreground or target-centric prompt strategies while neglecting the integration of background prompts.

Notably, the background prompting has proven effective in suppressing interference and enhancing target discrimination in various vision tasks, achieving significant improvements in medical segmentation [63], open-vocabulary detection [64], few-shot detection [65], and hyperspectral target detection [66], [67].

However, such background prompting mechanisms remain unexplored in MOT, particularly for challenging UAV-based multispectral scenarios. Unlike the relatively stable backgrounds in medical segmentation or detection tasks, UAV platforms exhibit rapid ego-motion and frequent attitude changes, resulting in highly dynamic BC that varies significantly across frames [68]. Furthermore, targets in UAV scenes are often small objects with limited pixels, leading to sparse feature representations. These domain-specific characteristics render sole reliance on target-aware positive prompts insufficient for maintaining robust tracking, as the model lacks sufficient discriminative cues from the target alone under severe background interference. Therefore, we propose PNPM-Net that leverages background prompting jointly with target-aware positive prompts. Specifically, we explicitly model background as structured negative prompts and integrate them with positive prompts via our CAPE and DIPM modules, transforming background from passive interference into active discriminative guidance.

III. Proposed Methods

A. Overview

The overall architecture of the proposed PNPM-Net is illustrated in Fig. 1. The framework consists of three main components: data preprocessing layer, positive–negative prompt mining (PNPM) backbone, and box head.

At the input stage, PNPM-Net adopts a multitemporal template pool with three uniformly distributed historical frames: $T_0 = I_0$ (long-term anchor), $T_M = I_{\lfloor k/3 \rfloor}$ (mid-term state), and $T_L = I_{\lfloor 2k/3 \rfloor}$ (recent changes), where $k = t-1$. These templates and the search region are processed by the data preprocessing

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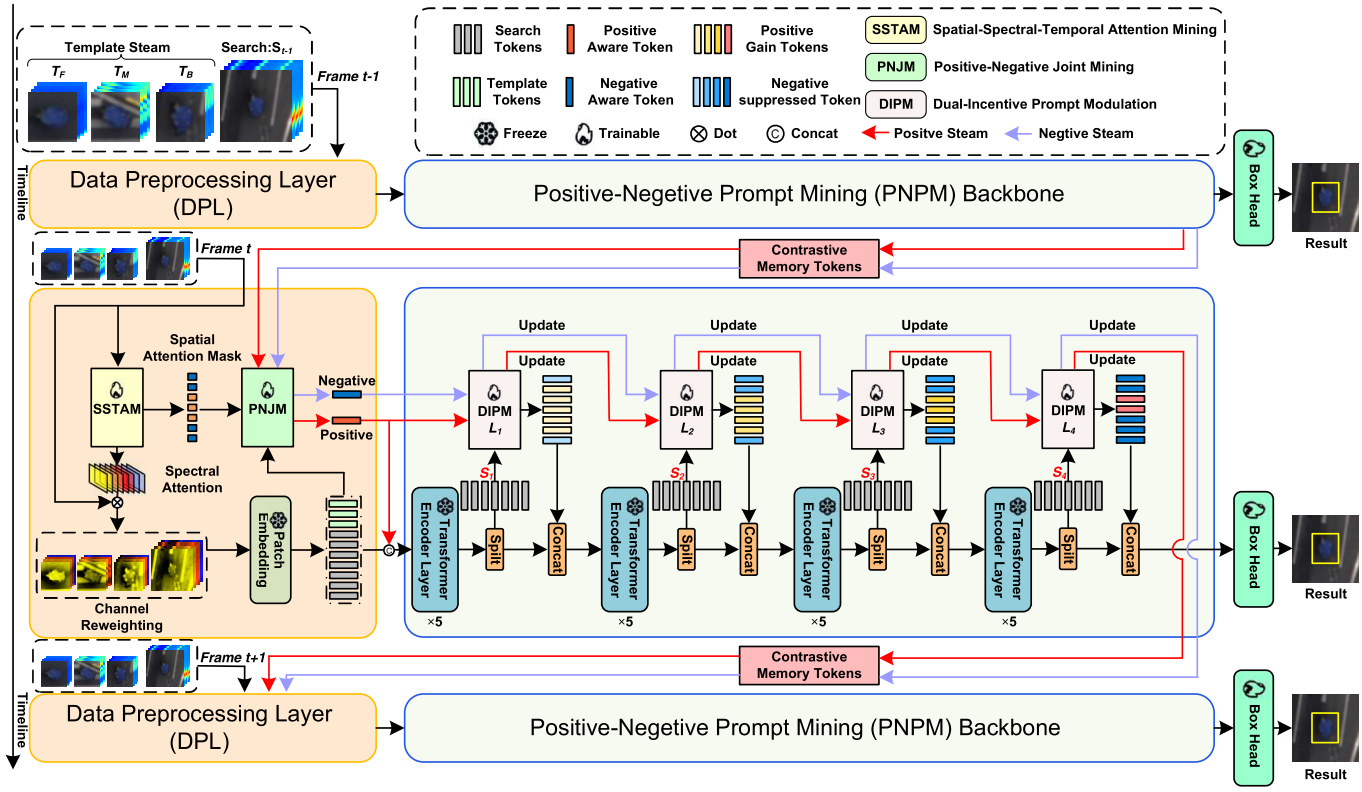


Fig. 1. Architecture of PNPM-Net. PNPM-Net consists of three main components: data preprocessing layer, PNPM backbone, and box head.

layer, in which SSTAM extracts spectral–spatial–temporal attention, and PNJM generates structured positive and negative prompts.

The prompts are then injected into the Transformer of the PNPM backbone via DIPM modules at multiple layers. DIPM performs parallel modulation, where positive prompts enhance target responses while negative prompts suppress confusing background regions. Finally, the box head predicts bounding boxes from the enhanced features.

B. Spatial–Spectral–Temporal Attention Mining

The motivation behind SSTAM is to jointly optimize the temporally aware spectral weights and spatial masks for both templates and the search region through an adaptive learning mechanism, thereby achieving effective fusion across spatial, spectral, and temporal dimensions. Specifically, SSTAM enhances spectral discriminability and SNR by suppressing noise-sensitive bands, while its multitemporal fusion captures target appearance evolution for improved adaptability. These optimizations serve as a foundation for the PNJM module: the enhanced SNR ensures purer feature representations for prompt generation, while the generated spatial masks provide essential constraints to prevent target contamination.

Specifically, SSTAM consists of two complementary branches. First, the spectral–temporal attention mining (SpeTAM) submodule dynamically adjusts spectral bands using a multitemporal template pool, thereby enhancing the distinguishability between target and background while generating template temporal weights to guide template feature fusion. Second, the spatial–temporal attention mining (SpaTAM) submodule uses intertemplate temporal weights to reshape spatial–temporal features. It then correlates these with search region features to generate a spatial attention mask for target–background segmentation. The architecture of the SSTAM module is illustrated in Fig. 2.

1) Spectral–Temporal Attention Mining: To effectively fuse temporal appearance evolution and enhance spectral discriminability, we propose the SpeTAM module. Operating on the multitemporal template pool T_t ($t \in \{0, M, L\}$), SpeTAM adaptively models spectral consistency across historical templates to generate temporally aware, channelwise spectral attention weights. These weights are then used to reweight features of both the template and search regions, thereby emphasizing spectrally stable and target-discriminative bands while suppressing noise-prone or inconsistent channels.

Specifically, each template T_t is first processed by a shared lightweight encoder, denoted as CRA (Conv–ReLU–AvgPooling), to produce a channel descriptor that integrates both spatial and spectral contextual information as follows:

$$e_t = \text{AvgPool}\left(\text{ReLU}\left(\mathbf{W}_c * T_t\right)\right) \in \mathbb{R}^C \tag{1}$$

where $\mathbf{W}_c \in \mathbb{R}^{3\times3\times C}$ denotes the weights of a standard 2-D convolutional layer with a spatial kernel size of 3×3 and C input channels. The operation maps these C input channels to C output channels, thereby modeling interchannel interactions while preserving the spectral dimension, AvgPool denotes

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the average pooling operation, C represents the number of template spectral channels, and e_t denotes the spatial–spectral fusion vector of template T_t, which compactly encodes spatially aggregated and interchannel contextual information for subsequent temporal fusion.

To enhance intertemplate semantic consistency, we stack the three-channel vector e_t as $\mathbf{E} = [e_0, e_M, e_L]^\top$ and compute a self-attention similarity matrix within a template attention scoring (TAS) block as follows:

$$\mathbf{A} = \text{Softmax}\left(\frac{\mathbf{E}\mathbf{E}^\top}{\sqrt{C}}\right) \in \mathbb{R}^{3\times 3} \tag{2}$$

where $\mathbf{A}$ represents the normalized similarity matrix among the three temporal templates, and each entry $A_{t,t'}$ quantifies how much template T_t should attend to template $T_{t'}$ based on their spectral-channel similarity. Each channel of templates representation is then weighted by the similarity matrix $\mathbf{A}$ for contextually enhancing and mapped to a scalar temporal confidence score $s_t \in \mathbb{R}$ via a lightweight MLP as follows:

$$s_t = \text{MLP}\left(\sum_{t'} A_{t,t'}\, e_{t'}\right), \quad t, t' \in \{0, M, L\} \tag{3}$$

where s_t reflects the reliability of template T_t by measuring its spectral consistency within the multitemporal context, and serves as the basis for computing adaptive temporal attention weights in the subsequent fusion step.

Finally, s_0, s_M, and s_L are normalized via L2 normalization to yield temporal attention weights $\alpha_t = s_t/(s_0^2+s_M^2+s_L^2+\epsilon)^{1/2}$ for $t \in \{0, M, L\}$, where ϵ is a small constant for numerical stability. Following the computation of the temporal attention weights $\boldsymbol{\alpha} = [\alpha_0, \alpha_M, \alpha_L]^\top \in \mathbb{R}^3$, the SpeTAM module fuses the channel descriptors from all temporal templates and applies a channelwise softmax function to generate a shared spectral attention weight vector $\mathbf{w}_{\text{spe}}$ as follows:

$$\mathbf{w}_{\text{spe}} = \text{Softmax}\left(\sum_{t\in\{0,M,L\}} \alpha_t\, e_t\right) \in \mathbb{R}^C. \tag{4}$$

Then, $\mathbf{w}_{\text{spe}}$ is broadcast across the spatial dimensions and used to reweight the original MS features of the search region S and each template T_t via channelwise multiplication to get spectral reweighting template $\tilde{T}_t$ and spectral reweighting search $\tilde{S}$ as follows:

$$\begin{cases} \tilde{S} & = \mathbf{w}_{\text{spe}} \otimes \mathbf{1}_{H_s\times W_s} \odot S \\ \tilde{T}_t & = \mathbf{w}_{\text{spe}} \otimes \mathbf{1}_{H_t\times W_t} \odot T_t \end{cases} \tag{5}$$

where $\otimes$ denotes the outer product, $\odot$ denotes the elementwise multiplication, and $\mathbf{1}$ is an all-one spatial tensor. $H_s \times W_s$ and $H_t \times W_t$ denote the spatial resolutions of the search region and template, respectively. The outputs $\tilde{S}$ and $\tilde{T}_t$ represent the reweighted search region and reweighted templates, respectively. Notably, SpeTAM employs a serial coupling design of temporal consistency-guided spectral weighting, reflecting the physical prior that spectral discriminability varies dynamically over time in UAV scenarios. This mechanism ensures that only spectrally stable bands are emphasized, while inconsistent templates receive lower weights via (3), naturally suppressing noise propagation to the spectral dimension.

2) Spatial–Temporal Attention Mining: To explicitly separate the target from similar background regions by exploiting cross-frame spatial consistency, we introduce the SpaTAM module.

Specifically, the three reweighted templates $\tilde{T}_0, \tilde{T}_M, \tilde{T}_L$ are first fused along the temporal dimension using the temporal attention weights $\boldsymbol{\alpha} = [\alpha_0, \alpha_M, \alpha_L]^\top$ as $\bar{T} = \sum_{t\in\{0,M,L\}} \alpha_t \tilde{T}_t$, where $\bar{T}$ is the temporally fused template. Subsequently, both $\bar{T}$ and $\tilde{S}$ are first reduced to single-channel grayscale maps by summing over the channel dimension. Then, these features are processed by the patch–AvgPooling–normalization (PAN) block, which partitions each map into nonoverlapping 16×16 patches, computes the average intensity within each patch, and applies layer normalization to obtain the spatial structure tokens for the template region T_{spa} and search region T_{spa} as follows:

$$\begin{cases} T_{\text{spa}} & = \text{PAN}\left(\sum_{c=1}^{C} \bar{T}_c\right) \\ S_{\text{spa}} & = \text{PAN}\left(\sum_{c=1}^{C} \tilde{S}_c\right). \end{cases} \tag{6}$$

To capture local structural consistency while emphasizing the central regions of the template, we compute a Gaussian-weighted sliding-window similarity map between the search and template spatial maps S_{spa} and T_{spa}. Specifically, for each spatial location p in the search map, a window of size $H_t \times W_t$ that matches the spatial extent of the fused template $\bar{T}$ is extracted. The dissimilarity map $E(p)$ at p is defined as a Gaussian-weighted sum of pixelwise squared differences as follows:

$$E(i,j) = \sum_{u=1}^{H_t}\sum_{v=1}^{W_t} G(u,v)\cdot\left(S_{\text{spa}}(i+u, j+v) - T_{\text{spa}}(u,v)\right)^2 \tag{7}$$

where $G(u,v) = \exp(-((u-u_c)^2+(v-v_c)^2)/2)$ is a 2-D isotropic Gaussian kernel centered at $(u_c, v_c) = ((H_t+1)/2, (W_t+1)/2)$. The dissimilarity map E is then converted into a foreground similarity map $\mathbf{M}_{\text{sim}}$ via a negative exponential transformation as $\mathbf{M}_{\text{sim}}(i,j) = \exp(-E(i,j)/2)$.

Finally, the similarity map $\mathbf{M}_{\text{sim}}$ is flattened and thresholded using its mean value to produce a binary foreground mask. The background attention mask $\mathbf{M}_{\text{bg}}$ is then obtained by logical inversion as follows:

$$\mathbf{M}_{\text{bg}} = 1 - \mathbb{I}(\mathbf{M}_{\text{sim}} > Avg(\mathbf{M}_{\text{sim}})) \in \{0, 1\} \tag{8}$$

where $\mathbb{I}(\cdot)$ denotes the binary function, $Avg(\cdot)$ denotes the mean function, $\mathbf{M}_{\text{bg}} = 1$ indicates regions identified as background and $\mathbf{M}_{\text{bg}} = 0$ corresponds to the target. Since $\mathbf{M}_{\text{bg}}$ is derived from multitemporal fused templates (8) rather than a single frame, the semantic stability of the mask is enhanced at the source, reducing the likelihood of initial classification errors. $\mathbf{M}_{\text{bg}}$ is subsequently used in the PNJM module to identify background tokens, enabling the structured negative prompt generation while suppressing the interference from potential target regions.

C. Positive–Negative Joint Mining

Building upon the refined feature representations and spatial constraints established by SSTAM, the motivation behind

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the PNJM module is to explicitly decouple the target from spectrally similar BC by constructing structured, complementary target-aware and background-aware prompt signals. As illustrated in Fig. 3 Specifically, the spectral reweighted multitemporal templates $\{\tilde{T}_0, \tilde{T}_M, \tilde{T}_L\}$ and the reweighted search region $\tilde{S}$ are first projected into token sequences via patch embedding. The computation of the embedded tokens is formulated as follows:

$$\begin{cases} Z_t &= \mathcal{E}\left(\tilde{T}_t\right) \in \mathbb{R}^{1\times 144\times 512}, \quad t \in \{0, M, L\} \\ X &= \mathcal{E}\left(\tilde{S}\right) \in \mathbb{R}^{1\times 576\times 512} \end{cases} \tag{9}$$

where $\mathcal{E}(\cdot)$ denotes the patch embedding operation, Z_t represents the token sequence of the tth temporal template, and X denotes the token sequence of the search region. The three template token sequences are then concatenated along the spatial dimension to form a unified template representation as $Z = [Z_0, Z_M, Z_L]$.

Next, the positive prompt is generated through a memory-augmented temporal fusion process. The three template token sequences Z_t ($t \in \{0, M, L\}$) are first fused using the temporal attention weights $\boldsymbol{\alpha} = [\alpha_0, \alpha_M, \alpha_L]^\top$ from SpeTAM and then globally aggregated over the token dimension as follows:

$$P_{\text{curr}} = \text{AvgPool}\left(\sum_{t\in\{0,M,L\}} \alpha_t \cdot Z_t\right) \in \mathbb{R}^{1\times 1\times 512}. \tag{10}$$

Concurrently, a negative prompt is generated through a background-aware similarity optimization process. First, the current positive prompt $P^t_{\text{pos},0}$ is transposed and multiplied with the search tokens X to compute tokenwise similarity scores. The background attention score $\mathbf{a}_{\text{bg}}$ is then obtained by inverting the softmax similarity and masking out nonbackground regions using the background attention mask $\mathbf{M}_{\text{bg}}$ from SpaTAM as follows:

$$\mathbf{a}_{\text{bg}} = \left(\mathbf{1} - \text{Softmax}\left(X \otimes \left(P^t_{\text{pos},0}\right)^\top\right)\right) \odot \mathbf{M}_{\text{bg}} \in \mathbb{R}^{1\times 576\times 1} \tag{11}$$

where $\mathbf{1}$ denotes an all-one vector. This operation directs the negative prompts to focus specifically on the background regions. Due to the similarity term assigning near-zero weights to regions highly similar to the positive prompt, the target pixels mistakenly included in $\mathbf{M}_{\text{bg}}$ are mathematically suppressed during the weighted aggregation. The resulting background attention scores $\mathbf{a}_{\text{bg}}$ are then used to reweight the search tokens, and the nonzero background components are aggregated through tokenwise average pooling to produce the current negative descriptor as follows:

$$N_{\text{curr}} = \text{AvgPool}\left(\mathbf{a}_{\text{bg}} \odot X\right) \in \mathbb{R}^{1\times 1\times 512}. \tag{12}$$

Finally, the current descriptors P_{curr} and N_{curr} are concatenated with the memory positive and negative tokens from the previous frame, $P^{t-1}_{\text{pos},4}$ and $P^{t-1}_{\text{neg},4}$ (both $\in \mathbb{R}^{1\times 1\times 512}$), along the channel dimension. These concatenated features are then refined by a lightweight MLP to generate the initial positive and negative prompt tokens for the current frame, $P^t_{\text{pos},0}$ and $P^t_{\text{neg},0}$, respectively, as follows:

$$\begin{cases} P^t_{\text{pos},0} &= \text{MLP}\left(\left[P_{\text{curr}},\, P^{t-1}_{\text{pos},4}\right]\right) \in \mathbb{R}^{1\times 1\times 512} \\ P^t_{\text{neg},0} &= \text{MLP}\left(\left[N_{\text{curr}},\, P^{t-1}_{\text{neg},4}\right]\right) \in \mathbb{R}^{1\times 1\times 512}. \end{cases} \tag{13}$$

Furthermore, this memory-enhanced fusion acts as a temporal low-pass filter. If abnormal noise or mask errors occur in a specific frame, the stable historical memory tokens provide a prior that prevents drastic fluctuations, gradually correcting biases in subsequent frames.

D. Dual-Incentive Prompt Modulation

The motivation of DIPM is to jointly modulate search features using target-aware positive prompts and background-aware negative prompts, dynamically enhancing target-consistent responses and suppressing background regions. As illustrated in Fig. 4.

Within the PNPM backbone, DIPM is inserted periodically every $5n$th encoder layer of Transformer for $n = 1, 2, 3$, and 4. At each insertion point, the input token sequence $\mathbf{U}$ is split into its constituent components according to the original layout as $\mathbf{U} = [P^t_{\text{pos},0}, Z, X]$. The search tokens X, the positive prompt $P^t_{\text{pos},n}$, and the negative prompt $P^t_{\text{neg},n}$ from the previous DIPM stage are then jointly fed into the current DIPM module for dual-incentive modulation. To balance modulation effectiveness and computational efficiency while achieving feature integration between positive–negative prompts and search tokens, DIPM adopts a tokenwise squeeze-and-excitation strategy. Specifically, all input tokens are compressed via a lightweight LMG block (LayerNorm + MLP + GELU) to reduce the channel dimension from 512 to 256 to enhance the attention efficacy of the prompts toward the search tokens. The detailed computation is as follows:

$$\begin{cases} \hat{x} &= \text{LMG}_{\text{dw}}(X) \in \mathbb{R}^{B\times 576\times 256} \\ \hat{p}_+ &= \text{LMG}_{\text{dw}}\left(P^t_{\text{pos},n}\right) \in \mathbb{R}^{B\times 1\times 256} \\ \hat{p}_- &= \text{LMG}_{\text{dw}}\left(P^t_{\text{neg},n}\right) \in \mathbb{R}^{B\times 1\times 256} \end{cases} \tag{14}$$

where $\hat{x}$ denotes the compressed search tokens, LMG_{dw} denotes the compression operation of the LMG Block, and $\hat{p}_+$ and $\hat{p}_-$ denote the compressed positive and negative prompts, respectively. Subsequently, the compressed positive prompt $\hat{p}_+$ and negative prompt $\hat{p}_-$ modulate the compressed search tokens $\hat{X}$ through two parallel cross-attention modules. The positive prompt enhances target-consistent features, while the negative prompt suppresses background confusing regions. The specific computations are as follows:

$$\begin{cases} \tilde{p}_+ &= \text{CrossAttn_Feat}\left(\mathbf{Q} = \hat{p}_+, \mathbf{K} = \hat{x}, \mathbf{V} = \hat{x}\right) \\ &\quad \in \mathbb{R}^{B\times 1\times 256} \\ \tilde{m}_+ &= \text{CrossAttn_Map}\left(\mathbf{Q} = \hat{p}_+, \mathbf{K} = \hat{x}\right) \in \mathbb{R}^{B\times 1\times 576} \\ \tilde{p}_- &= \text{CrossAttn_Feat}\left(\mathbf{Q} = \hat{p}_-, \mathbf{K} = \hat{x}, \mathbf{V} = \hat{x}\right) \\ &\quad \in \mathbb{R}^{B\times 1\times 256} \\ \tilde{m}_- &= \text{CrossAttn_Map}\left(\mathbf{Q} = \hat{p}_-, \mathbf{K} = \hat{x}\right) \in \mathbb{R}^{B\times 1\times 576} \end{cases} \tag{15}$$

where CrossAttn_Feat(·) denotes the standard dot-product cross attention, while CrossAttn_Map(·) refers to the intermediate attention weight matrix, $\tilde{p}_+$ and $\tilde{p}_-$ are the modulated positive and negative prompt features, respectively, and $\tilde{m}_+$ and $\tilde{m}_-$ represent their corresponding attention maps over the search region. Subsequently, the positive and negative attention maps $\tilde{m}_+$ and $\tilde{m}_-$ are broadcast along the channel dimension

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to match X and then fused to modulate the original search tokens, yielding the modulated output X_{out} as follows:

$$X_{\text{out}} = \left(\mathbf{1} + \tilde{m}_{+} - \tilde{m}_{-}\right) \odot X \in \mathbb{R}^{B\times 576\times 512} \tag{16}$$

where $\mathbf{1}$ denotes an all-ones vector of the same length as $\tilde{m}_{+}$ and $\tilde{m}_{-}$. Simultaneously, the compressed positive and negative prompt tokens $\hat{p}_{+}$ and $\hat{p}_{-}$ are passed through an excitation LMG block to restore their channel dimension to 512 and generate updated prompt tokens. These tokens are then combined with the input prompts via residual connections to produce the optimized positive and negative prompts for the next DIPM stage as follows:

$$\begin{cases} P^{t}_{\text{pos},n+1} &= P^{t}_{\text{pos},n} + \text{LMG}_{\text{up}}\left(\tilde{p}_{+}\right) \in \mathbb{R}^{B\times 1\times 512} \\ P^{t}_{\text{neg},n+1} &= P^{t}_{\text{neg},n} + \text{LMG}_{\text{up}}\left(\tilde{p}_{-}\right) \in \mathbb{R}^{B\times 1\times 512} \end{cases} \tag{17}$$

where LMG_{up} denotes the excitation operation of LMG block. Following modulation, the modulated search tokens $\mathbf{X}_{\text{out}}$ and the updated positive prompt $\mathbf{P}^{t}_{\text{pos},n}$ are combined with their original tokens from the input of the current DIPM stage via a learnable gating parameter $\theta \in [0, 1]$ to achieve optimization of the corresponding features as follows:

$$\begin{cases} \tilde{P}^{t}_{\text{pos},0} &= \theta_n \cdot P^{t}_{\text{pos},0} + (1-\theta_n)\cdot P^{t}_{\text{pos},n+1} \in \mathbb{R}^{B\times 1\times 512} \\ \tilde{X} &= \theta_n \cdot X + (1-\theta_n)\cdot X_{\text{out}} \in \mathbb{R}^{B\times 576\times 512}. \end{cases} \tag{18}$$

Here, θ_n denotes the layer-specific gating parameter for the nth DIPM module. Unlike a globally shared scalar, each DIPM stage maintains an independent learnable θ_n, allowing the network to adaptively balance the original and modulated features according to the semantic characteristics at different depths. This ensures that shallow layers can preserve detailed textures while deep layers focus on high-level discriminative cues, optimizing feature fusion throughout the backbone. Finally, the updated positive prompt $\tilde{P}^{t}_{\text{pos},0}$, the template tokens Z, and the optimized search tokens $\tilde{X}$ are concatenated along the token dimension to reconstruct the full sequence for the next Transformer layer as $\mathbf{U}^{\text{next}} = [\tilde{P}^{t}_{\text{pos}}, Z, \tilde{X}]$, where $\mathbf{U}^{\text{next}}$ denotes the reassembled token sequence that will be fed into the subsequent encoder layer of the PPIN backbone.

E. Head and Loss

In this study, we employ a center-based prediction head that predicts three dense maps over the search region: a classification score map $\mathbf{C} \in \mathbb{R}^{H_p\times W_p}$, a center offset map $\mathbf{O} \in \mathbb{R}^{2\times H_p\times W_p}$, and a scale change map $\mathbf{R} \in \mathbb{R}^{2\times H_p\times W_p}$, where $H_p = H_s/16$ and $W_p = W_s/16$ denote the height and width of the output feature grid, respectively. The classification map $\mathbf{C}$ encodes the likelihood that the target is centered at each spatial location. The offset map $\mathbf{O}$ provides subpixel refinement for the center coordinates, and the scale map $\mathbf{R}$ captures normalized adjustments to the width and height of the target to account for SVs. All values in $\mathbf{O}$ and $\mathbf{R}$ are normalized to the range $[0, 1]$. The final target bounding box is obtained by selecting the location with the maximum score in $\mathbf{C}$, applying the corresponding offset from $\mathbf{O}$ to refine the center, and scaling the base dimensions according to $\mathbf{R}$.

Meanwhile, the training objective of PNPM-Net is defined by a composite loss function that jointly optimizes classification and localization performance. As formulated below, the total loss L comprises three terms aligned with the network's prediction heads

$$L = L_{\text{cls}} + \lambda_1 L_{\text{IoU}} + \lambda_2 L_{\text{L1}} \tag{19}$$

where L_{cls}, L_{IoU}, and L_{L1} correspond to the focal loss [69], IoU loss [70], and L1 loss, respectively. The weighting factors λ_1 and λ_2 are introduced to harmonize the optimization of different tasks.

IV. Experiments and Discussions

A. Experimental Setting

1) Dataset: In our experiment, PNPM-Net was trained and tested on the MUST [26] dataset, which is the first large-scale MS tracking benchmark specifically designed for UAV platforms. The dataset contains a total of 43 000 frames, corresponding to a video duration of approximately 8600 s. It includes 160 sequences for the training set and 90 sequences for the testing set. Each frame contains eight spectral bands covering the visible to near-infrared range, captured from altitudes of 20–250 m. The dataset includes frame-level bounding box annotations and is labeled with 12 challenge attributes: BC, camera motion (CM), fast motion (FM), full occlusion (FOC), illumination variation (IV), low resolution (LR), MB, out-of-view (OV), partial occlusion (POC), similar color (SC), similar object background (SOB), and SV. These attributes comprehensively reflect the realistic challenges in UAV-based object tracking.

2) Implementation Details: The proposed PNPM-Net is implemented in PyTorch 1.13.1 using Python 3.8 and is trained on a machine equipped with an Intel i9-12900K processor, 64 GB of RAM, and an NVIDIA RTX 3090 GPU with 24 GB of VRAM. Specifically, we employ HiViT [71] as the backbone architecture, initialized with weights pretrained on large-scale RGB tracking benchmarks, including GOT-10k [72], LaSOT [73], COCO [74], and TrackingNet [75]. Notably, during the fine-tuning phase on the MUST dataset, the backbone parameters are kept frozen to focus the optimization on the proposed MS interaction modules. Optimization for the trainable parameters is performed using the AdamW optimizer. For this fine-tuning process, the initial learning rate is set to 1×10^{-4}. To enhance training stability and prevent over-convergence, a drop path rate of 0.3 and a gradient clipping norm of 0.001 are applied. The model is trained for a total of 30 epochs with a batch size of 2. The learning rate is decayed by a factor of 0.1 starting from the 10th epoch. Input images are resized such that template patches are 192×192 pixels and search regions are 384×384 pixels.

3) Evaluation Metrics: This study assesses the performance of the proposed tracker using three standard metrics: success rate, precision, and area under the curve (AUC) [30]. The success rate indicates the fraction of frames in which the predicted bounding box overlaps sufficiently with the ground truth, as measured by an intersection-over-union (IoU) threshold. Precision, on the other hand, reflects the proportion

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TABLE I

OVERALL PERFORMANCE OF VARIOUS TRACKERS WITH THE BEST AND SECOND-BEST VALUES HIGHLIGHTED IN RED AND ITALICIZED AND BLUE AND UNDERLINED FONTS, RESPECTIVELY

Method	Year	AUC	DP@20	FPS	MOP
MHT [31]	2020	0.359	0.497	2.76	CPU
BAE-Net [84]	2020	0.380	0.535	2.11	GPU
TransT [77]	2021	0.458	0.635	*50.2*	GPU
MixFormer [78]	2022	0.496	0.732	33	GPU
OSTrack [79]	2022	0.468	0.643	33.6	GPU
SEE-Net [35]	2023	0.418	0.592	10.96	GPU
SiamBAG [83]	2023	0.403	0.546	12.56	GPU
DaSPPNet [85]	2024	0.563	0.756	21.2	GPU
EVPTrack [80]	2024	0.517	0.731	24.2	GPU
MMF-Net [86]	2024	0.470	0.604	15.86	GPU
ODTrack [81]	2024	0.525	0.764	19.14	GPU
HCSMP [36]	2025	0.536	0.757	17.42	GPU
HDSP [37]	2025	0.515	0.657	7.88	GPU
MambaLCT [82]	2025	0.602	0.754	19.16	GPU
SSTtrack [38]	2025	0.600	0.778	15.52	GPU
UNTrack [27]	2025	0.606	0.807	21.6	GPU
PNPM-Net	Ours	*0.718*	*0.881*	23.67	GPU

of frames where the center of the predicted bounding box lies within a specified distance from the ground-truth center. By convention, precision is reported at a 20-pixel distance threshold, denoted as DP@20 [30]. AUC serves as the primary criterion for ranking trackers, and all evaluations follow a one-pass protocol.

In addition to accuracy, we evaluate efficiency using FPS, FLOPs, and Params. Frames per second (FPS) [30] quantifies the inference speed and real-time capability, calculated based on the average single-frame inference time across all challenge sequences. Floating point operations (FLOPs) [30] measure the computational complexity per forward pass, and Params [30] denotes the total number of learnable parameters, reflecting the model size. Together, these metrics provide a holistic view of the effectiveness and practicality.

B. Comparison to SOTA Trackers

In this section, we conduct a comprehensive comparison between the proposed PNPM-Net and 16 SOTA tracking methods to evaluate both qualitative and quantitative performances. The SOTA methods and publication venues are summarized in Table I. Among these baselines, Transformer tracking (TransT) [76], MixFormer [77], OSTrack [78], EVPTrack [79], ODtrack [80], and MambaLCT [81] are designed for RGB-based visual tracking, while the remaining approaches, including MHT [30], SEE-Net [34], SiamBAG [82], BAE-Net [83], DaSPPNet [84], MMF-Net [85], HCSMP [35], HDSP [36], SSTtrack [37], and UNTrack [26] are MS trackers.

In the experiments, we standardized the training protocol across all compared methods. And all deep learning-based SOTA methods were trained and tested on a uniform platform powered by NVIDIA RTX 3090 GPUs. Meanwhile, baseline models and the proposed PNPM-Net were retrained on the MUST dataset for 30 epochs with a fixed learning rate of 1×10^{-4}. To align the input data with the architectural requirements of different trackers, specific preprocessing strategies

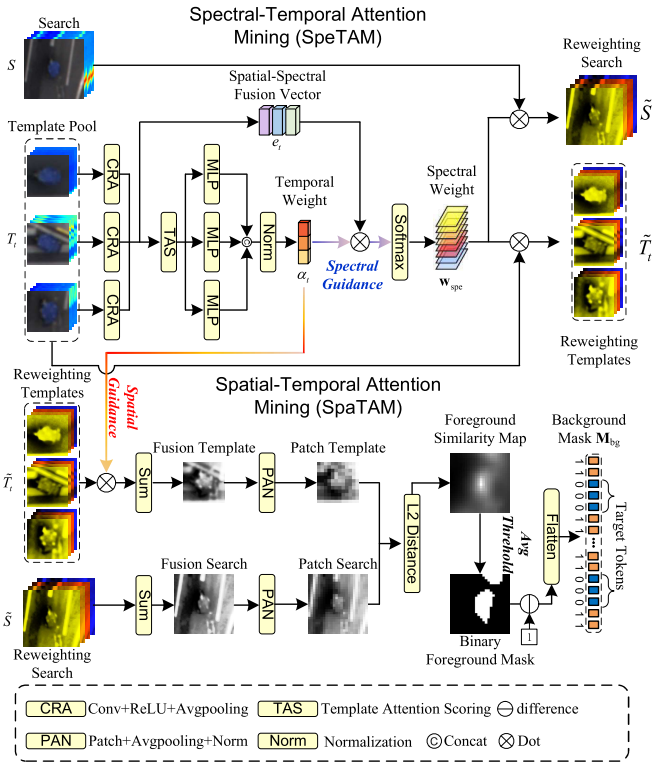


Fig. 2. Architecture of the SSTAM module. SSTAM consists of two complementary submodules: SpeTAM and SpaTAM. SpeTAM employs the temporal attention weights to guide spectral channel reweighting for discriminative band enhancement. SpaTAM leverages cross-frame temporal consistency to reshape spatial features and generate a background-aware attention mask.

were applied. For the MS trackers, the original eight-band MUST data were adapted via interpolation to match the specific number of spectral bands required by each model. For the RGB-based trackers, pseudo-color images were constructed by selecting bands [0, 4, 7] from the eight-band cube at approximately equal intervals to simulate standard RGB inputs. Furthermore, the data augmentation strategy was kept consistent across all methods to objectively and accurately reflect the practical capabilities of each tracking paradigm.

1) Overall Comparison to SOTA Trackers: Fig. 5 and Table I summarize the quantitative and qualitative comparison between PNPM-Net and the 16 SOTA trackers. As shown in Table I, PNPM-Net achieves the highest AUC of 0.718 and DP@20 of 0.881. It outperforms the best comparing MS tracker in comparison, UNTrack [26], by 18.5% in AUC and 9.2% in DP@20. Compared with the top-performing RGB-based tracker, MambaLCT [81], it improves AUC by 19.3% and DP@20 by 16.8%. In terms of efficiency, PNPM-Net runs at 23.67 FPS on a GPU. It is faster than most MS trackers and achieves much higher accuracy than the top-two fastest trackers like OSTrack [78] and TransT [76]. The success and precision plots in Fig. 5 further show its consistent superiority in success rate and precision, highlighting the potential of PNPM-Net in UAV-based MS tracking. Furthermore, Table I reveals that among SOTA methods, the trackers with prompt learning, such as UNTrack [26], SSTtrack [37], and HCSMP [35], consistently outperform other approaches,

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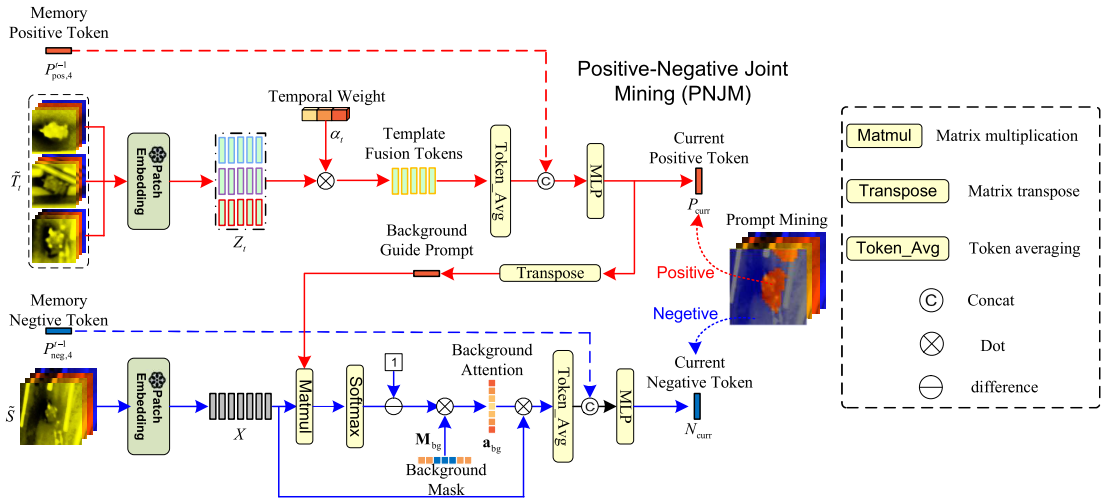


Fig. 3. Architecture of the PNJM module. PNJM jointly processes the spectrally reweighted multitemporal templates and the current search region to explicitly mine background characteristics into structured negative prompt tokens, transforming background interference into active discriminative knowledge alongside positive prompts.

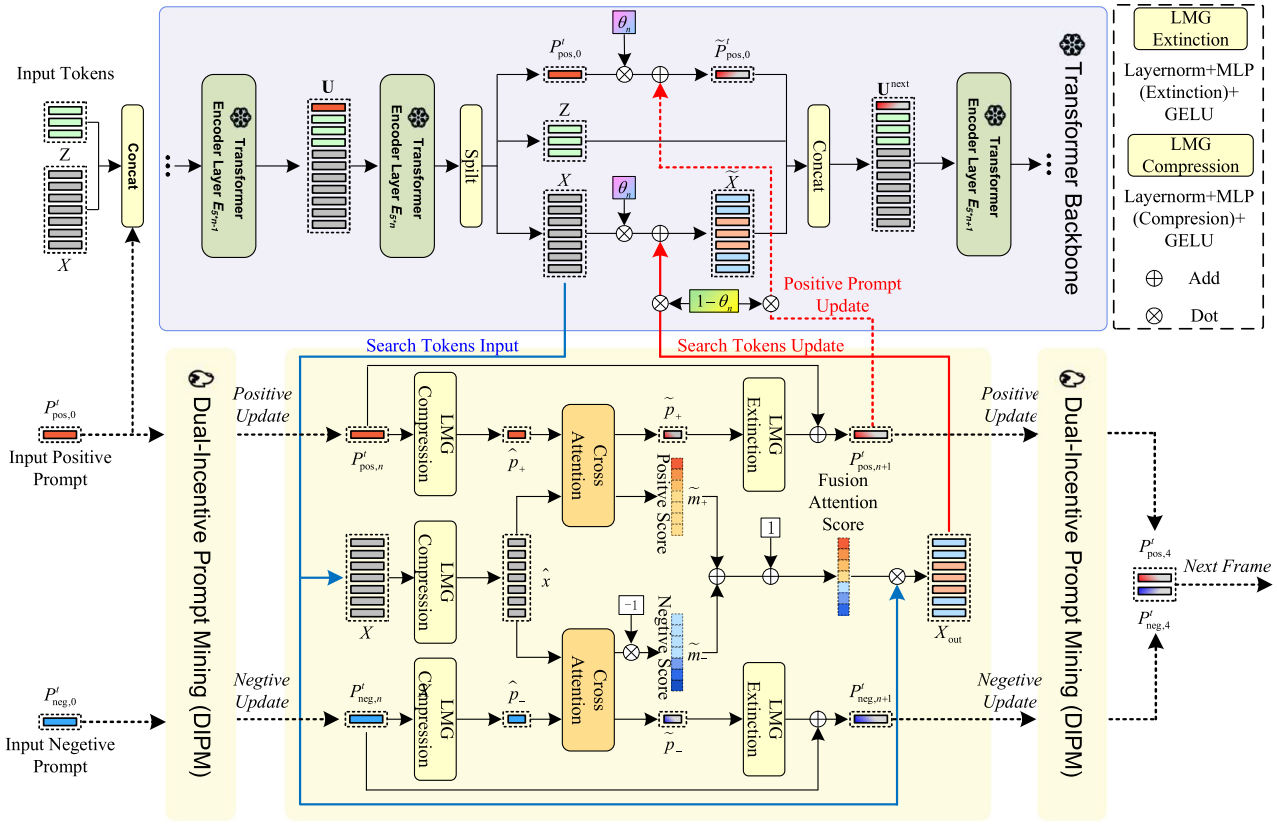


Fig. 4. Architecture of the DIPM module. The DIPM module is periodically inserted at multiple intermediate layers of the Transformer backbone to dynamically refine search-region features through cooperative modulation by positive and negative prompts.

demonstrating the effectiveness of prompt learning in leveraging MS contextual information. Building upon this paradigm, PNPM-Net surpasses existing prompt-based designs by modeling background interference as structured negative prompts and jointly mining them with target-aware positive prompts.

2) Attribute-Based Comparison to SOTA Trackers: To comprehensively evaluate the robustness of PNPM-Net on the MUST dataset under diverse challenges, we compare it against ten MS trackers and six RGB trackers across 12 distinct attributes. Table II provides a detailed comparison of the performance of these trackers on each attribute. According to the comparative results, PNPM-Net ranks in the top two in 11 of 12 evaluated attributes. Notably, it exhibits exceptional performance on the CM, SC, and SOB challenges. Specifically, it surpasses the second-best SOTA tracker by 21.2%, 15.5%, and 22.9% in AUC, and by 10.7%, 6.5%, and 13.5% in DP@20,

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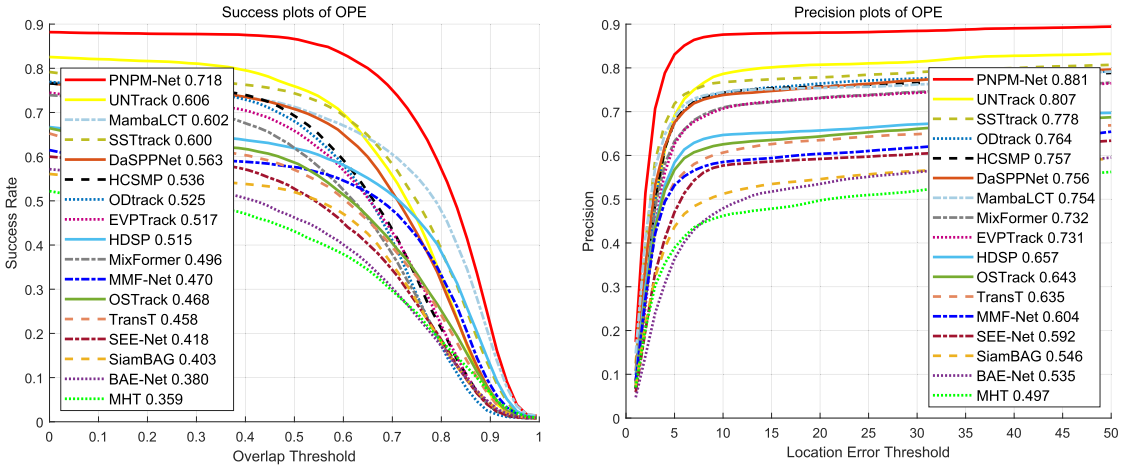


Fig. 5. Comparison of success rate and precision against SOTA MS and RGB trackers on the MUST dataset.

TABLE II

PERFORMANCE COMPARISON (AUC/DP@20) OF DIFFERENT TRACKERS ON MULTIPLE ATTRIBUTES. THE BEST (1ST) AND SECOND-BEST (2ND) RESULTS ARE HIGHLIGHTED IN RED ITALIC AND BLUE UNDERLINE, RESPECTIVELY

Method	BC	CM	FM	FOC	IV	LR	MB	OV	POC	SC	SOB	SV
MHT [31]	0.327/0.491	0.329/0.422	0.334/0.439	0.206/0.288	0.350/0.487	0.181/0.306	0.291/0.396	0.400/0.537	0.371/0.515	0.305/0.434	0.417/0.561	0.361/0.521
TransT [77]	0.355/0.532	0.476/0.642	0.454/0.612	0.276/0.406	0.391/0.564	0.260/0.429	0.448/0.598	0.405/0.525	0.441/0.608	0.376/0.557	0.500/0.657	0.391/0.560
MixFormer [78]	0.392/0.639	0.516/0.733	0.493/0.712	0.332/0.503	0.398/0.622	0.319/0.570	0.555/0.783	0.409/0.584	0.467/0.687	0.411/0.660	0.514/0.742	0.458/0.701
OSTrack [79]	0.345/0.497	0.478/0.653	0.459/0.620	0.351/0.518	0.481/0.684	0.279/0.466	0.524/0.741	0.436/0.600	0.468/0.642	0.425/0.627	0.470/0.611	0.414/0.583
SEE-Net [35]	0.382/0.572	0.379/0.520	0.381/0.532	0.283/0.406	0.437/0.620	0.203/0.350	0.369/0.509	0.323/0.423	0.417/0.594	0.359/0.521	0.455/0.610	0.386/0.591
SiamBAG [83]	0.341/0.480	0.373/0.485	0.369/0.486	0.255/0.349	0.390/0.519	0.227/0.342	0.344/0.455	0.323/0.426	0.391/0.532	0.326/0.448	0.456/0.614	0.427/0.581
BAE-Net [84]	0.386/0.579	0.262/0.353	0.311/0.424	0.305/0.458	0.367/0.537	0.200/0.341	0.269/0.398	0.352/0.469	0.446/0.625	0.289/0.410	0.426/0.615	0.373/0.610
DaSPPNet [85]	0.475/0.673	0.572/0.760	0.560/0.751	0.442/0.631	0.528/0.729	0.405/0.611	0.623/0.820	0.453/0.600	0.546/0.730	0.525/0.736	0.555/0.719	0.484/0.678
EVPTrack [80]	0.447/0.665	0.541/0.759	0.511/0.718	0.353/0.542	0.477/0.677	0.358/0.603	0.529/0.726	0.459/0.626	0.505/0.708	0.477/0.699	0.517/0.707	0.443/0.657
MMF-Net [86]	0.446/0.603	0.421/0.527	0.445/0.567	0.370/0.506	0.484/0.634	0.252/0.371	0.407/0.509	0.460/0.609	0.483/0.624	0.408/0.536	0.511/0.647	0.452/0.583
ODtrack [81]	0.433/0.650	0.572/0.816	0.550/0.791	0.375/0.593	0.477/0.734	0.366/0.635	0.653/0.905	0.476/0.697	0.500/0.721	0.495/0.751	0.504/0.739	0.466/0.717
HCSMP [36]	0.435/0.638	0.567/0.799	0.529/0.749	0.421/0.643	0.506/0.710	0.385/0.639	0.624/0.895	0.400/0.530	0.529/0.743	0.511/0.764	0.512/0.710	0.443/0.648
HDSP [37]	0.446/0.570	0.486/0.635	0.476/0.616	0.436/0.592	0.547/0.677	0.364/0.528	0.439/0.565	0.454/0.530	0.504/0.627	0.498/0.648	0.480/0.599	0.485/0.642
MambaLCT [82]	0.598/0.758	0.570/0.734	0.552/0.700	0.517/0.697	0.665/0.826	0.481/0.661	0.453/0.578	*0.601*/*0.709*	0.638/0.786	0.595/0.761	0.543/0.659	0.548/0.705
SSTtrack [38]	0.525/0.710	0.596/0.759	0.601/0.774	0.523/0.716	0.593/0.797	0.439/0.635	0.583/0.734	0.561/0.702	0.610/0.787	0.562/0.758	0.576/0.727	0.546/0.738
UNTrack [27]	0.543/0.728	0.609/0.823	0.578/0.773	0.524/0.739	0.655/*0.867*	0.508/0.734	0.632/0.871	0.413/0.528	0.609/0.802	0.618/0.850	0.550/0.708	0.546/0.739
PNPM-Net	*0.632*/*0.799*	*0.738*/*0.911*	*0.707*/*0.873*	*0.599*/*0.769*	*0.700*/0.861	*0.616*/*0.805*	*0.776*/*0.952*	0.557/0.660	*0.701*/*0.854*	*0.714*/*0.905*	*0.708*/*0.842*	*0.611*/*0.768*

respectively. This advantage stems from the joint operation of the CAPE and DIPM modules. In CM scenarios, where abrupt UAV viewpoint changes cause rapid target deformation or displacement, the SSTAM submodule in CAPE performs dynamic spectral reweighting across frames.

In CM scenarios, abrupt changes in the UAV viewpoint often induce rapid target deformation or significant displacement. To address this, the SSTAM module introduces a cross-frame dynamic spectral reweighting mechanism. This mechanism adaptively enhances spectrally stable and discriminatively strong bands over time while suppressing channels affected by noise or exhibiting interframe inconsistency. Such temporally coherent spectral reshaping significantly improves spectral separability and enables robust generation of target-aware positive and background-aware negative prompts. Meanwhile, these negative prompts actively suppress dynamic BC induced by ego-motion, complementing the target enhancement provided by positive prompts. As a result, the tracking system maintains high stability even under severe target appearance variations. In SC and SOB scenarios, the target exhibits high visual similarity to its surroundings or is surrounded by semantically similar distractors. PNPM-Net leverages the PNJM module to construct structured negative prompts modeling background interference, specifically identifying spectrally similar regions via the background attention mask. Subsequently, the DIPM module applies dual-incentive modulation, where negative prompts actively suppress search features correlating with background profiles while positive prompts enhance target responses. This contrastive mechanism prevents drift onto similar objects by inhibiting background responses rather than solely enhancing target features, ensuring accurate localization under strong spectral interference. Therefore, the results validate the effectiveness of PNPM-Net in complex, challenging scenarios.

Nevertheless, our proposed method exhibits certain limitations, particularly in the OV challenge. Specifically, PNPM-Net achieves an AUC of 0.557 in OV scenarios, which is lower than 0.601 of MambaLCT [81]. This limitation

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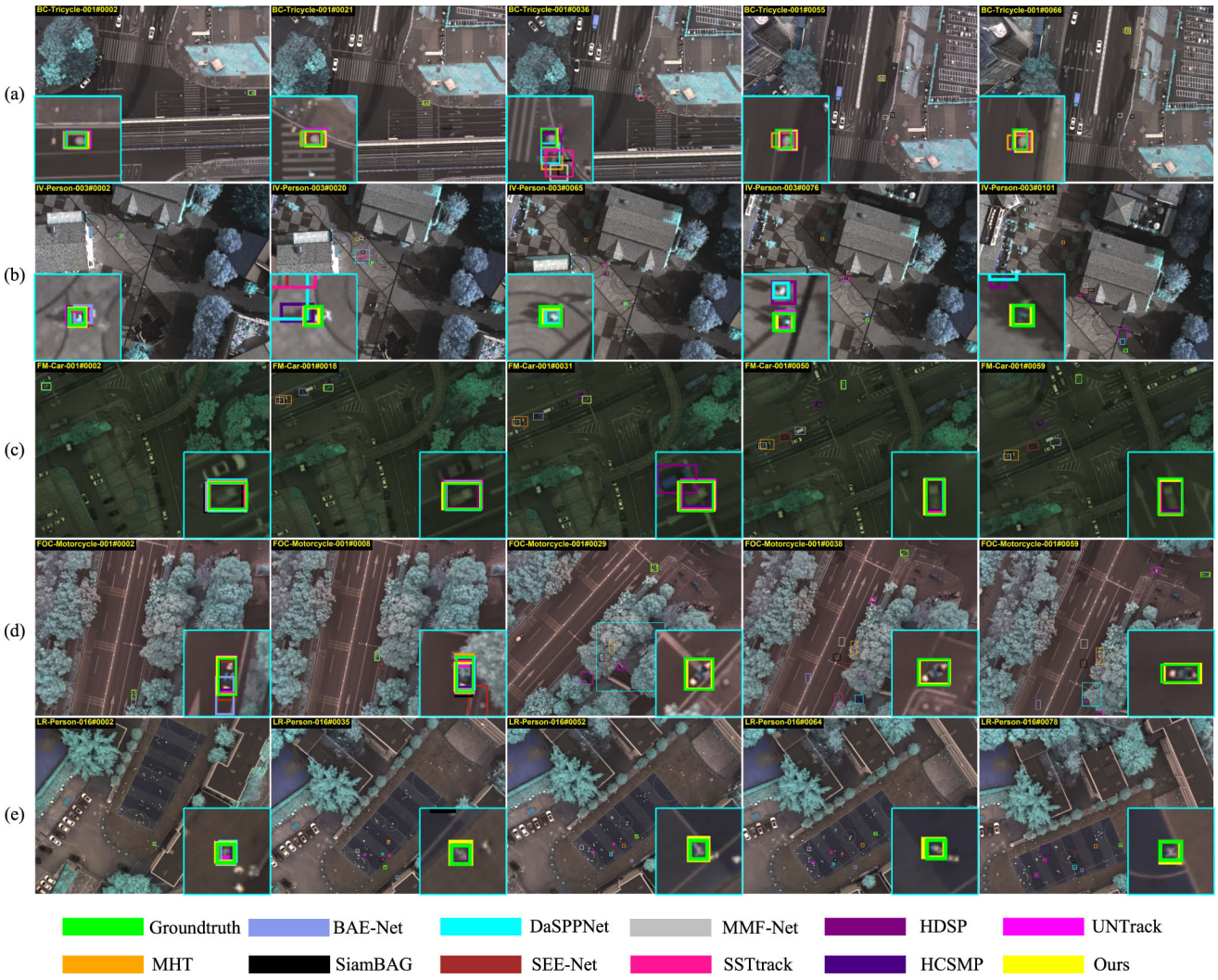


Fig. 6. Visualization of tracking results on five video sequences from the MUST dataset. Due to the small target size in UAV aerial views, we zoomed-in view the target regions (shown in the lower right or lower left corners) to facilitate clearer observation of tracking bounding boxes. Each row corresponds to a different sequence. (a) *BC-Tricycle-001*, (b) *IV-Person-003*, (c) *FM-Car-001*, (d) *FOC-Motorcycle-001*, and (e) *LR-Person-016*. Different colored bounding boxes represent different tracking methods. The ground truth is shown in green, while results from comparative methods and our proposed approach are displayed in various colors.

primarily stems from the negative prompt generation mechanism's reliance on target–background contrast. Within PNJM, background attention scores are derived by inverting similarity to the positive prompt. While the multitemporal template pool effectively handles short-term occlusion, prolonged target absence renders historical templates stale and uncorrelated with the search region. Consequently, the positive prompt loses discriminative power, yielding noisy background scores and nondiscriminative negative prompts. Upon target reappearance, this accumulated noise hinders the rapid reconstruction of reliable cues, thereby compromising overall robustness compared with state-space models that maintain long-term context independently.

3) Visualization Tracking Comparison: To demonstrate the tracking performance of PNPM-Net against SOTA MS trackers, we conducted a qualitative comparison of five tracking sequences. These sequences were carefully selected based on criteria of covering high-frequency challenge attributes and diverse target types to ensure the representativeness and systematic nature of the qualitative analysis. Specifically, the sequences include *BC-Tricycle-001* (POC, FM, CM, IV, BC, and SC), *FM-Car-001* (FM, CM, and MB), *FOC-Motorcycle-001* (POC, FOC, CM, SV, and LR), *IV-Person-003* (SOB, FM, CM, IV, LR, and SC), and *LR-Person-016* (SOB, FM, CM, BC, LR, and SC). The results are shown in Fig. 6. Bounding boxes in different colors represent results from different tracking algorithms, with the yellow bounding box denoting the proposed PNPM-Net. These sequences primarily involve challenges such as BC, SC, or SOB, which are used to validate the advantage of the proposed method in handling complex background interference scenarios.

As shown in Fig. 6, in the *BC-Tricycle-001 sequence*, the presence of substantial BC on the sidewalk causes MMF-Net [85] and SiamBAG [82] to misidentify road textures as the target, leading to tracking failure. In the *IV-Person-003* sequence, the target pedestrian is surrounded by multiple

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TABLE III

RUNNING COSTS COMPARISON RESULTS INCLUDING FPS, PARAMS, AND FLOPS. THE BEST (1ST) AND SECOND-BEST (2ND) RESULTS ARE HIGHLIGHTED IN RED ITALIC AND BLUE UNDERLINE, RESPECTIVELY

Method	PNPM-Net	UNTrack	SSTtrack	HDSP	HCSMP	MMF-Net	DaSPPNet	BAE-Net	SiamBAG	SEE-Net	MHT
FPS	*23.67*	21.6	15.52	7.88	17.42	15.86	21.2	2.11	12.56	10.96	2.76
Params(M)	136.3	144.2	27.2	253.2	125.2	45.34	181.4	*17.7*	90.25	54.7	—
FLOPs(T)	*0.235*	0.256	1.64	0.267	0.263	0.353	0.276	1.48	0.274	0.343	—

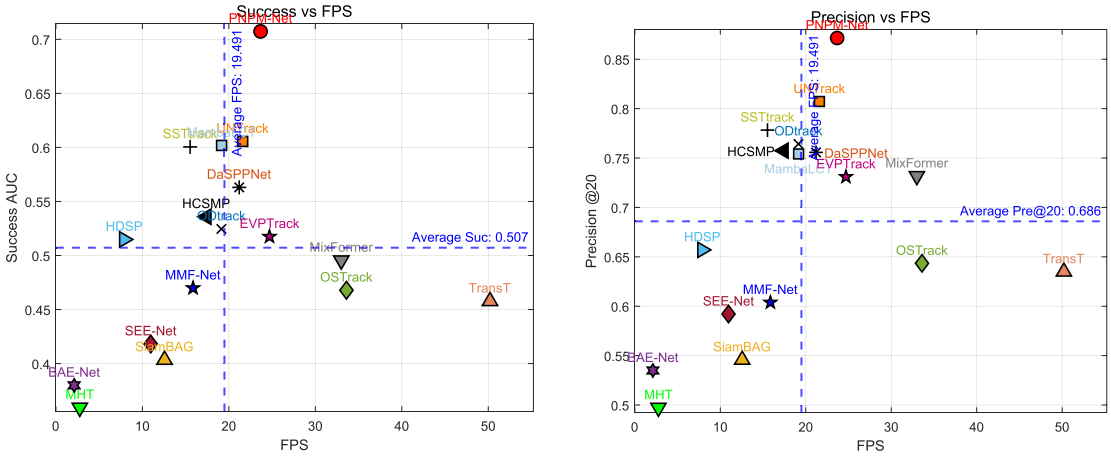


Fig. 7. Performance–efficiency comparison with MS and RGB trackers on the MUST dataset. (left) Success versus FPS. (right) Precision versus FPS.

distractors with similar appearance and spectral characteristics, causing DaSPPNet [84] and HDSP [36] to suffer severe drift at frame 76. In *FOC-Motorcycle-001*, the target is continuously occluded by trees starting from frame 8, and most trackers lose the target due to the absence of an effective dynamic temporal template modeling mechanism. In *LR-Person-016*, the combination of LR and densely distributed similar objects poses a significant challenge for the majority of SOTA trackers, limiting their ability to maintain stable tracking. In contrast, our proposed PNPM-Net consistently achieves accurate and stable tracking across all these challenging scenarios. This performance robustness is attributed to two key innovations. First, the SSTAM module enables the dynamic spectral reshaping between targets and background, while PNJM jointly mines target-aware positive and background-aware negative prompts to enhance the distinctions between the object and the background. Second, the DIPM module leverages these positive and negative prompts to dynamically guide feature modulation via bidirectional incentives. Therefore, PNPM-Net effectively resists interference in complex UAV-based MSI scenarios.

4) Running Costs Comparison: Real-time performance and computational efficiency are critical requirements in visual tracking tasks. To comprehensively evaluate the practical deployment capability of our proposed method, we conduct a systematic comparison against SOTA MS trackers across three key metrics, including computational complexity (FLOPs), model parameter count (Params), and inference speed (FPS). All experiments are performed under a unified environment on an Ubuntu platform equipped with an NVIDIA RTX 3090 GPU to ensure fair comparison. As shown in Table III, among all MS trackers, BAE-Net [83] achieves the best parameter count; however, its multichannel grouped computation mechanism leads to substantially increased computational overhead, resulting in FLOPs as high as 1.48T and thus limited efficiency. Meanwhile, HCSMP [35] and HDSP [36] rely on layerwise prompt learning strategies and require parallel input of multiple false-color images, which not only imposes the memory burden but also degrades inference speed, which yield FPS values of 17.42 and 7.88, respectively.

In contrast, the proposed PNPM-Net achieves the highest inference speed and the lowest FLOPs among all compared MS trackers. This advantage stems from its DIPM module, which employs a segmentwise prompt learning strategy instead of layerwise prompting. This design dramatically reduces model parameters while simultaneously delivering significant performance gains. Furthermore, Fig. 7 illustrates the overall performance with FPS of all methods on the MUST dataset. As shown in Fig. 7, PNPM-Net stands out by maintaining a higher frame rate than the average FPS while achieving superior localization accuracy and robustness compared with most SOTA approaches. These results validate the potential of PNPM-Net as an efficient and practical tracker for real-world applications.

C. Ablation Study

In this section, we conduct a series of ablation studies on the MUST dataset to demonstrate the effectiveness of the proposed method.

1) Evaluation of Each Component: PNPM-Net consists of three core components: SSTAM, PNJM, and DIPM. We adopt the network architecture with the three core modules removed as the baseline model (denoted by "B"). Within the data preprocessing layer, the SSTAM module first integrates

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TABLE IV

ABLATION EXPERIMENTAL RESULTS (AUC) ON DIFFERENT CONFIGURATIONS OF 12 ATTRIBUTE CHALLENGES, WITH THE BEST AND SECOND-BEST VALUES ARE HIGHLIGHTED IN RED AND ITALICIZED AND BLUE AND UNDERLINED FONTS, RESPECTIVELY

Method	BC	CM	FM	FOC	IV	LR	MB	OV	POC	SC	SOB	SV
B	0.428	0.568	0.541	0.377	0.470	0.362	0.650	0.452	0.498	0.491	0.501	0.452
B+S	0.572	0.677	0.669	0.476	0.589	0.592	0.667	0.542	0.524	0.613	0.629	0.542
B+S+P	0.614	0.705	0.662	0.535	0.693	0.584	0.727	*0.582*	0.552	0.655	0.664	0.535
B+S+P+D	*0.632*	*0.738*	*0.707*	*0.599*	*0.700*	*0.616*	*0.776*	0.557	*0.701*	*0.714*	*0.708*	*0.611*

TABLE V

ABLATION EXPERIMENTAL RESULTS (AUC AND DP@20) ON DIFFERENT CONFIGURATIONS. Δ INDICATES THE AMOUNT OF VARIATION. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	AUC	DP@20	Δ(AUC)	Δ(DP@20)
B	0.515	0.718	-	-
B+S	0.633	0.842	0.108	0.078
B+S+P	0.682	0.851	0.157	0.087
B+S+P+D	**0.718**	**0.881**	**+0.203**	**+0.163**

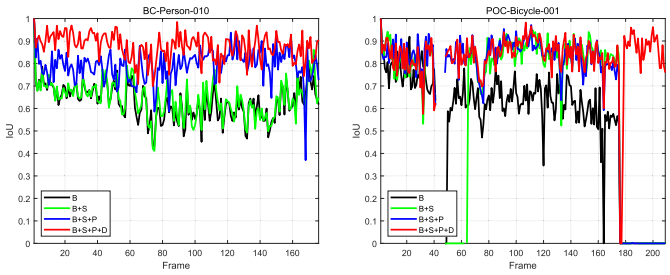


Fig. 8. Ablation study of PNPM-Net with different configurations on IoU per frame for two video sequences (i.e., *BC-Person-010* and *POC-Bicycle-001*).

spatial, spectral, and temporal cues to adaptively generate spectral weights and background attention masks (denoted by "S"). Subsequently, the PNJM module leverages the contextual features produced by SSTAM to jointly generate a target-aware positive prompt and a background-aware negative prompt (denoted by "P"). Finally, the DIPM module (denoted by "D") dynamically injects these dual-excitation prompts into multiple layers of the backbone network, performing bidirectional modulation of the search region features through positive and negative prompts.

We first conducted attribute-based ablation experiments of each component. The AUC scores of different configurations are summarized in Table IV. On the MUST dataset, which defines 12 challenging attributes, our full model (B+S+P+D) outperforms the ablation baselines on 11 of them. Under the OV scenario, when the target remains completely absent from the field of view for an extended period, the historical appearance information stored in the multitemporal template pool gradually becomes outdated. This causes the DIPM module to lose discriminative power due to the lack of reliable responses in the search region, ultimately degrading tracking robustness. Meanwhile, the overall quantitative ablation results are shown in Table V. Compared with the baseline, progressively adding the SSTAM, PNJM, and DIPM modules consistently improves performance. Also, the full model achieves gains of + 0.203 in AUC and + 0.163 in DP@20 over the baseline, demonstrating the effectiveness of each component in enhancing both tracking robustness and accuracy.

As shown in Fig. 8, we further present the IOU curves of two highly challenging sequences, including *BC-Person-010* (POC, SOB, and BC) and *POC-Bicycle-001* (POC, FOC, FM, CM, BC, LR, and SC) to visually validate the effectiveness of our proposed modules. In *BC-Person-010*, due to the absence of the PNJM and DIPM blocks, the ablation models (e.g., B+S and B) struggle to effectively discriminate the target from background interference in complex scenarios, resulting in a significant drop in their IoU compared with the full PNPM-Net (B+S+P+D). Similarly, in *POC-Bicycle-001*, starting from frame 175, the target bicycle is obstructed and surrounded by similar tree distractors. All baseline models suffer from noticeable drift or complete tracking failure, whereas only the full PNPM-Net (B+S+P+D) maintains a consistently high IoU. This demonstrates that PNJM effectively mines discriminative positive–negative prompt pairs, while DIPM leverages the negative prompts to actively suppress background regions spectrally similar to the target, enabling stable and accurate tracking even in highly cluttered scenes.

TABLE VI

EVALUATION OF THE PERFORMANCE OF PNPM-NET BASED ON THE NUMBER OF TEMPLATES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Templates	1	2	3	4	5
AUC	0.652	0.696	**0.718**	0.715	0.709
DP@20	0.844	0.879	0.881	**0.884**	0.878
Params(M)	**119.8**	125.4	136.3	157.6	181.2
FPS	**29.6**	26.2	23.67	17.8	13.2

2) Number of Input Templates in Network: The PNPM-Net utilizes a multitemporal template pool to capture appearance evolution across different time scales, thereby enhancing robustness against complex scenarios encountered in UAV-based MS tracking. Therefore, the number of input templates directly influences the diversity and stability of the learned positive–negative prompts, which in turn affects both tracking accuracy and computational efficiency. Therefore, we construct the ablation study about the configurations with varying numbers of reference templates (from 1 to 5) to evaluate their impact on the performance of PNPM-Net, and the results are shown in Table VI. In all configurations, the selected templates are uniformly distributed across historical frames.

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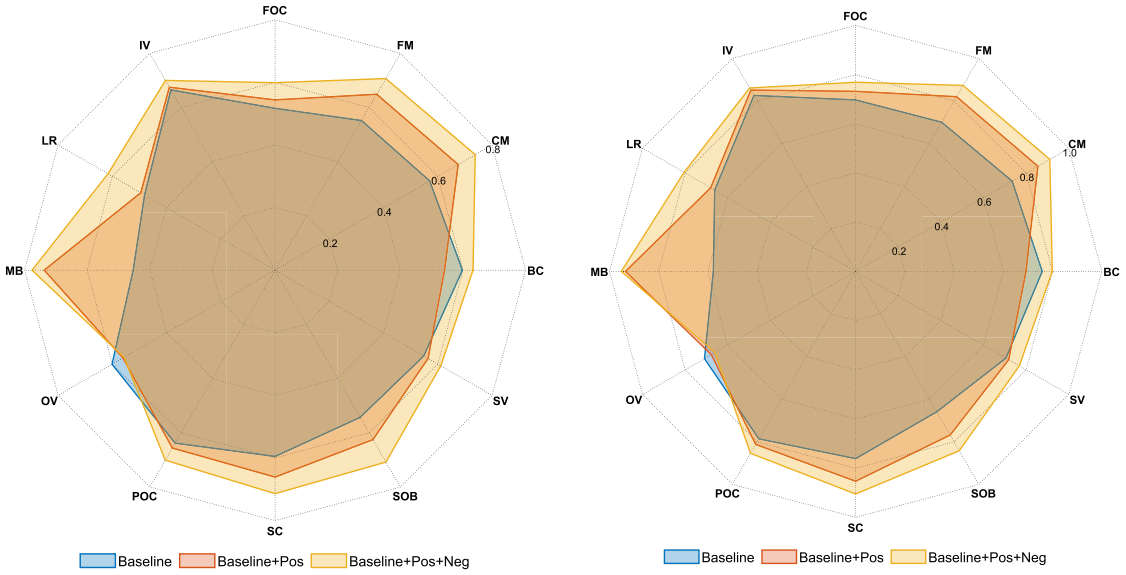


Fig. 9. Performance comparison of different configurations across challenge attributes in the MUST dataset. (a) AUC radar chart. (b) DP@20 radar chart.

TABLE VII
ABLATION STUDY OF POSITIVE AND NEGATIVE PROMPT USAGE. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Positive Prompt	Negetive Prompt	AUC	DP@20	FPS
×	×	0.602	0.754	**27.25**
✓	×	0.658	0.827	24.85
×	✓	0.681	0.855	24.79
✓	✓	**0.718**	**0.881**	23.67

When only a single template is used, the model achieves the highest inference speed (29.6 FPS) and the lowest parameter number (119.8M), but suffers from limited temporal coverage, resulting in suboptimal AUC (0.652) and DP@20 (0.844). Increasing the number of templates to 3 significantly improves tracking accuracy, yielding the best AUC of 0.718, while maintaining a competitive DP@20 of 0.881 and a real-time speed of 23.67 FPS. Further increasing the number of templates to four slightly boosts DP@20 to 0.884, but at the cost of reduced AUC (0.715), increased model size (157.6M), and slower inference (17.8 FPS). Using five templates leads to performance degradation in both AUC and DP@20, which is likely due to redundant or conflicting historical information that dilutes prompt quality. Considering the tradeoff among accuracy, model complexity, and speed, we select 3 templates as the default configuration in CAPE, which achieves the best overall performance while minimizing redundancy.

3) Ablation Study on DIPM Distribution: The DIPM module plays a critical role in PNPM-Net by dynamically refining search-region features through structured collaboration between positive and negative prompts. Consequently, the internal design and its deployment strategy across the network critically influence tracking robustness and computational efficiency. To verify the effectiveness of the dual-incentive learning mechanism of positive and negative prompts in the DIPM module, we first designed a systematic prompt learning ablation study. This experiment systematically evaluates the contributions of both types of prompts to tracking performance by selectively enabling or disabling the positive and negative prompts in the DIPM module. As shown in Table VII, the model achieves its best performance when both positive and negative prompts are enabled (AUC = 0.718 and DP@20 = 0.881), surpassing cases using only a single prompt or no prompt at all. Particularly noteworthy is that when negative prompts are used alone, the performance of the model (AUC = 0.681 and DP@20 = 0.855) outperforms the case with only positive prompts (AUC = 0.658 and DP@20 = 0.827). This indicates that in UAV MOT tasks, background information serving as negative prompts plays a crucial role in distinguishing the target from similar background interference.

Meanwhile, to investigate the performance differences of various configurations across different challenging attributes, we constructed radar charts for AUC and DP@20 under different settings, as illustrated in Fig. 9. These radar charts clearly demonstrate the performance of three configurations, including Baseline, Baseline+Pos (with only positive prompts open), and Baseline+Pos+Neg (with both positive and negative prompts open) across 12 challenging attributes. It can be observed from the chart that the Baseline+Pos+Neg configuration exhibits superior performance in the majority of attributes. In the LR scenario, due to the LR resulting in limited information from the target area, using only positive prompts has a limited ability to enhance feature representation. In contrast, by incorporating negative prompts, Baseline+Pos+Neg achieves a significant performance improvement. In the BC scenario, interference from the background causes the performance of the Baseline+Pos configuration to slightly underperform even the Baseline. However, Baseline+Pos+Neg introduces negative prompts to suppress background interference, effectively

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TABLE VIII

ABLATION EXPERIMENTAL RESULTS (AUC AND DP@20) UNDER DIFFERENT COMPRESSION DIMENSIONS IN DIPM. Δ INDICATES THE AMOUNT OF VARIATION RELATIVE TO THE BASELINE (512). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Channel	AUC	Δ(AUC)	DP@20	Δ(DP@20)	FPS
512	0.714	—	0.876	—	21.36
256	**0.718**	**+0.004**	**0.881**	**+0.005**	23.67
128	0.683	-0.031	0.864	-0.012	23.96
64	0.636	-0.078	0.784	-0.092	24.38
32	0.607	-0.107	0.758	-0.118	**24.92**

TABLE IX

EVALUATION OF THE PERFORMANCE OF OUR METHOD BASED ON THE NUMBER OF DIPM. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

DIPM Number	2	4	5	10	Each Layer
AUC	0.694	0.718	**0.719**	0.717	0.713
DP@20	0.854	**0.881**	0.879	0.874	0.869
Params(M)	**127.3**	136.3	142.7	176.4	262.3
FPS	**25.64**	23.67	21.16	17.75	12.15

improving tracking robustness under BC challenges. These results intuitively demonstrate the potential application value of modeling background information as structured negative prompts in UAV-based MOT tasks.

To investigate the impact of channel compression within DIPM, we conduct an ablation study on the compressed channel dimension used during cross-attention computation, as shown in Table VIII. It can be seen that reducing the channel dimension in DIPM from the original 512–256 not only lowers computational overhead but also improves the tracking performance, achieving the best AUC of 0.718(+0.004) and DP@20 of 0.881(+0.005), along with a notable FPS increase from 21.36 to 23.67. This phenomenon aligns with information bottleneck (IB) theory, which prioritizes retaining task-relevant information while discarding noise. In DIPM, the original 512-D features may contain redundant spectral noise or BC that dilutes attention focus. Moderate compression to 256 channels acts as a regularization bottleneck, filtering nondiscriminative variations and preserving salient target–background cues, thereby enhancing the SNR during cross-attention interaction for more precise modulation. Further compression (e.g., to 128 or lower) degrades accuracy despite yielding higher FPS. This indicates that while moderate compression facilitates effective feature interaction by removing redundancy, excessive compression impairs the representational capacity required for the discriminative modulation by causing significant information loss. Considering the tradeoff among accuracy, robustness, and efficiency, we adopt a compressed channel dimension of 256 as the default configuration in DIPM.

Furthermore, we evaluate the effect of DIPM placement by varying the number of DIPM modules inserted into the backbone on the MUST dataset (see Table IX). Performance improves when increasing DIPMs from 2 to 4: AUC rises from 0.694 to 0.718 and DP@20 increases from 0.854 to 0.881.

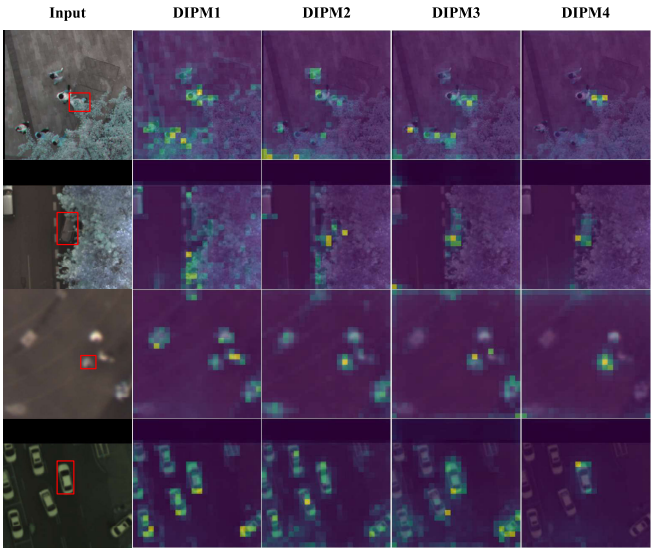


Fig. 10. Visualization of attention maps of PNPM-Net under different stages of DIPM block layers.

Beyond 4 modules, further increasing the number of DIPMs yields only marginal improvements in tracking accuracy, while significantly degrading inference speed. Meanwhile, the diminishing returns in performance, combined with a sharp drop in FPS, demonstrate that excessive insertion of DIPM modules is inefficient and counterproductive. Also, we also constructed a visualization sequence of attention maps for the search region, including the original input and features from DIPM1 to DIPM4, as shown in Fig. 10. In early stages, such as DIPM1, the modulation effect of the dual prompts is weak, resulting in diffuse attention and a weak target response. As the number of DIPM layers increases, the complementary dual-incentive mechanism becomes more pronounced: positive prompts progressively amplify target-consistent responses while negative prompts actively suppress background interference. This bidirectional modulation sharpens the feature contrast beyond simple feature addition, allowing the model to focus more accurately on the true target and achieve stronger spatial discrimination. The visual progression from DIPM1 to DIPM4 clearly demonstrates how the structured cooperation between these prompts refines the attention distribution, validating the effectiveness of the proposed collaborative mechanism. Considering both accuracy and efficiency, we select four DIPM modules as the final configuration, which fully leverages the benefits of prompt-based modulation while maintaining a balance between computational cost and inference speed.

V. CONCLUSION

In this article, we propose PNPM-Net to address tracking robustness degradation caused by target feature sparsity under complex background interference in UAV-based MS scenarios. First, an SSTAM module is designed to mine adaptive spectral–spatial–temporal attention, extracting discriminative bands to mitigate spectral ambiguity under rapid ego-motion. Second, a PNJM module generates structured positive and negative prompts, transforming background interference into

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discriminative guidance to overcome foreground-centric limitations. Finally, a DIPM module injects these prompts for dual-incentive modulation, enhancing target responses while suppressing confusing regions to significantly improve target–background discriminability. Extensive experiments demonstrate the strong potential of PNPM-Net in the UAV-based MOT domain.

In future work, we plan to address the limitations in OV scenarios by integrating long-term memory mechanisms or robust redetection modules. Specifically, we aim to enhance long-term memory mechanisms by integrating temporal consistency constraints or redetection modules to better handle extended occlusion or target absence, thereby further enhancing the robustness of UAV-based MS tracking in complex dynamic environments.

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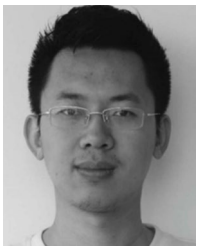
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